

Lemma: 倍角公式

let $\beta = \alpha$.

$$\begin{aligned}\sin(\alpha+\alpha) &= \sin \alpha \cos \alpha + \cos \alpha \sin \alpha & \therefore \sin 2\alpha &= 2\sin \alpha \cos \alpha. \\ &= \sin 2\alpha = 2\sin \alpha \cos \alpha.\end{aligned}$$

$$\begin{aligned}\cos(\alpha+\alpha) &= \cos \alpha \cos \alpha - \sin \alpha \sin \alpha \\ &= \cos^2 \alpha - \sin^2 \alpha\end{aligned}$$

$$\therefore \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha.$$

$$\tan(\alpha+\alpha) = \frac{\tan \alpha + \tan \alpha}{1 - \tan \alpha \tan \alpha}$$

$$= \frac{2\tan \alpha}{1 - \tan^2 \alpha}$$

$$\therefore \tan 2\alpha = \frac{2\tan \alpha}{1 - \tan^2 \alpha}.$$

$$\cos 2\alpha = 2\cos^2 \alpha - 1.$$

$$\Rightarrow \cos 2\alpha + 1 = 2\cos^2 \alpha.$$

$$= \frac{\cos 2\alpha + 1}{2} = \cos^2 \alpha.$$

$$\cos 2\alpha = 1 - 2\sin^2 \alpha \quad (\because \sin^2 + \cos^2 = 1).$$

$$= \cos^2 \frac{\alpha}{2} = \frac{\cos \alpha + 1}{2}$$

$$1 - \cos 2\alpha = 2\sin^2 \alpha.$$

$$\therefore \cos^2 \frac{\alpha}{2} = \frac{\cos \alpha + 1}{2}$$

$$\therefore \sin^2 \alpha = \frac{1 - \cos 2\alpha}{2} //$$

$$\Rightarrow \sin^2 \frac{\alpha}{2} = \frac{1 - \cos \alpha}{2} //$$